0G · PHYSICAL AI + PRIVATE COMPUTER · 15 SECONDS

A robot that cannot pass a checkpoint without 0G.

At every stop it pays an AI agent in 0G, which asks a private TEE model on 0G Private Computer what it sees. The answer and its request ID go into a signed episode on 0G Storage. If 0G is down it stops safely. It never fakes a decision.

private TEE inference | request ID per decision | paid in native 0G | episodes on 0G Storage | safe stop

0G IS THE BRAIN, NOT THE SPINAL CORD

The decision loop, at every checkpoint

1 · Drive
ROS 2 + Gazebo; navigation and safety stay on the robot

→

2 · Look
one camera frame, never stored

→

3 · Hire + pay
robot pays the agent owner in 0G; the transfer is checked on chain before 0G is called

→

4 · 0G PC
0gm-1.0-35b-a3b, private mode, TEE providers only

→

5 · Check
exact JSON, allowed action, confidence ≥ 0.6

→

6 · Act
clear the toys, report, hold or go home: bounded actions only

Model output is advisory. It picks one of five actions; it never touches motors, money or keys.

A REAL CALL, AS THE CUSTOMER SEES IT

"How it decided": every 0G call, in full

```
Sent through the 0G Router to 0gm-1.0-35b-a3b
in private mode, provider 0xf4FB…79B250
0G request ID  985498df-94a2-4e3d-9f1d-21ffe9f8a472
Time           24,464 ms
Billed to      the agent's own Router key · cap 0.5
USD/day
Answer  { "observation": "Colorful blocks on the
floor",
          "anomaly_type": "toys on floor",
          "confidence": 0.95,
          "recommended_action": "clear_and_continue" }
Checked  ✓ exact format  ✓ allowed action  ✓ 0.95 ≥ 0.6
Did      picked the toys up and put them in the bin
```

Privacy on demand

Private mode routes only to TEE providers; a model without TEE is never used for a private job. No silent downgrade.

Per-agent compute

Each agent calls 0G with its own Router key, created with a daily credit cap, allowed models and trust mode.

NATIVE 0G MONEY, ON 0G GALILEO

The robot has its own wallet and its own budget

Customer → robot
pays the robot's signed quote after confirming arrival (memo semmal:quote:<id>)

→

Robot → agent owner
one transfer per agent job (semmal:job:<id>), within the paid budget, never twice

→

Agent → 0G PC
its price covers its own capped 0G inference

→

Robot → owner / customer
the margin if done; the unspent rest refunded if not

Allowlisted recipients, per-payment and daily caps, idempotent. The only key is the robot's device key on the robot PC.

0G STORAGE: THE EPISODE LOG

Every job ends as a signed bundle, with a request ID per decision

```
episode bundle (canonical JSON)
  decisions[]   stop · providerRequestId · model
                trust mode · latency · verdict
  purchases[]   agent · 0G payment tx · job score
  actions[] · transitions[] · money · quote
  score         task-score-v1
  manifest_hash   sha256 · robot_signature

0G Storage root   0x45645ef9…7aefef
Arkiv index       task_score 100 · root · request IDs
```

Saved locally first

LOCAL_DURABLE → STORAGE_CONFIRMED → ARKIV_CONFIRMED, with retries and no duplicates.

Verified on read

The evidence view downloads with the Storage proof and checks the hash, the robot signature and the Arkiv link.

SAFE STOP: NO FAKE DECISIONS

If 0G can't answer, the robot doesn't guess

hold

no answer, late, invalid

The robot holds position and retries once with the same paid job.

SAFE_STOP

still no valid answer

It stops safely and records why. Nothing is invented. Tested: "0G fails twice → SAFE_STOP, nothing invented".

local

motor overcurrent

The safety supervisor stops the robot with no network and no AI; the incident is signed and queued until it reaches Storage and Arkiv.

DELIVERABLES

Both 0G tracks, checked

✓ 0G call inside the decision loop (vision at every stop)	`services/robot-master/src/mission.ts`
✓ Episode log on 0G Storage with a request ID per decision	`evidence.ts · decisions[].providerRequestId`
✓ Replayable simulation: deterministic Gazebo worlds	`scripts/robot/scenario.sh · demo video`
✓ Safe stop when 0G is unreachable; never a faked decision	`mission.test.ts`
✓ 0G PC in the main user flow, request IDs shown to the customer	`How it decided`
✓ Repo + setup, live link, addresses, team	`README · semmal.vercel.app`

WHAT BREAKS IF 0G IS REMOVED

The robot can't decide at a single checkpoint, so it safe-stops: no job done, no agent paid.

0G is the brain at each checkpoint, the money, and the memory, and the robot keeps the motion and the safety.

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